

2. Data Analytics I Create a Linear Regression Model using Python/R to predict trip prices using “taxi_trip_pricing” Dataset. Compare MSE of simple linear and multiple linear regression.

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# Import necessary libraries

import pandas as pd

import numpy as np

import matplotlib.pyplot as plt

from sklearn.model_selection import train_test_split

from sklearn.linear_model import LinearRegression

from sklearn.metrics import mean_squared_error


# Step 1: Load the dataset

df = pd.read_csv('taxi_trip_pricing.csv')


# Step 2: Data Preprocessing (check for missing values, remove irrelevant columns)

print(df.info())

print(df.head())


# Handle missing values (if any)

df.fillna(df.mean(), inplace=True)


# Step 3: Create Simple Linear Regression Model

# Assuming 'distance' is a feature to predict 'trip_price'

X_simple = df[['distance']] # Independent variable (simple feature)

y = df['trip_price']        # Dependent variable (target)


# Split the data into training and testing sets

X_train, X_test, y_train, y_test = train_test_split(X_simple, y, test_size=0.2, random_state=42)
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# Train the model

simple_regressor = LinearRegression()
simple_regressor.fit(X_train, y_train)

# Make predictions

y_pred_simple = simple_regressor.predict(X_test)

# Evaluate using Mean Squared Error (MSE)

mse_simple = mean_squared_error(y_test, y_pred_simple)
print(f"Simple Linear Regression MSE: {mse_simple}")

# Step 4: Create Multiple Linear Regression Model

# Assuming other features like 'distance', 'weather', 'time_of_day' are available in the dataset
# We'll include 'weather' and 'time_of_day' as additional features for multiple linear regression

X_multiple = df[['distance', 'weather', 'time_of_day']] # Multiple independent variables

# Split the data into training and testing sets

X_train, X_test, y_train, y_test = train_test_split(X_multiple, y, test_size=0.2, random_state=42)

# Train the model

multiple_regressor = LinearRegression()
multiple_regressor.fit(X_train, y_train)

# Make predictions

y_pred_multiple = multiple_regressor.predict(X_test)

# Evaluate using Mean Squared Error (MSE)

mse_multiple = mean_squared_error(y_test, y_pred_multiple)
print(f"Multiple Linear Regression MSE: {mse_multiple}")
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# Step 5: Compare MSE for both models

print("\nComparison of MSE for Simple and Multiple Linear Regression:")
print(f"Simple Linear Regression MSE: {mse_simple}")
print(f"Multiple Linear Regression MSE: {mse_multiple}")


# Optional: Plotting results for Simple Linear Regression
plt.scatter(X_test, y_test, color='blue')
plt.plot(X_test, y_pred_simple, color='red')
plt.title('Simple Linear Regression: Distance vs Trip Price')
plt.xlabel('Distance')
plt.ylabel('Trip Price')
plt.show()
```